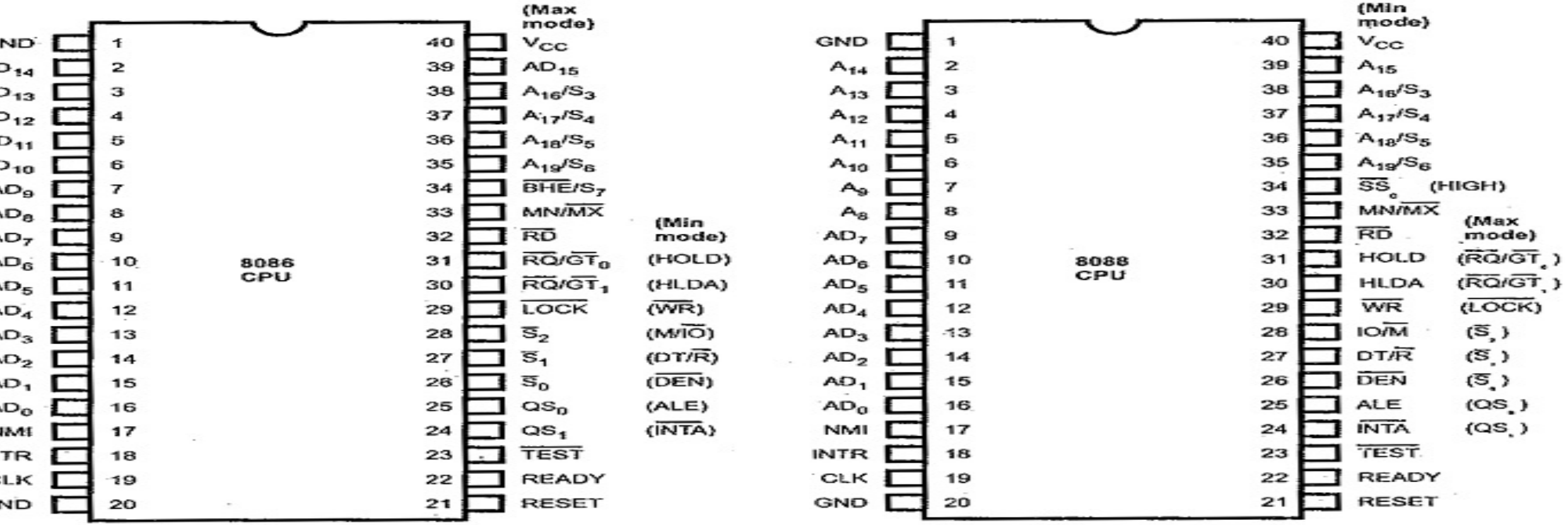
Hardware A

8088 vs 8086

- 8088 has 20 Address pins (A0-A19), 8 data pins (d0-d7)
- 8086 has 20 address pins (A0-A19), 16 data pins (d0-d15)



(a) Pin diagram of 8086 Fig. 10.1 (b) Pin diagram of 8088

MN/$\overline{MX}$

-
- 8088 has 2 modes
- Min mode→ single processor mode in the system , basic , control signals are generated (read, write , interrupt)
- Max mode → multiple processor (math) , complex , External controller (intel 8288)

- MN/$\overline{MX}$ → active low
- 0 max works
- 1 min works

Data bus

- Try to use min number of pins
- Multiplex： 8088 multiplex the lower 8 bits of the address bus and the data bus on AD0-AD7
- ALE： address latch enable to know the information is on address or data
- 0： data D0-D7
- 1： address A0-A7
- Demultiplex： separating the AD0-AD7

- LATCH： 74LS373 (most used latch) , works with ALE , to ensure the separation between Address and Data

Address bus

- LATCH ⇒ 74LS373 (most used latch) , works with ALE , to ensure the separation between Address and Data

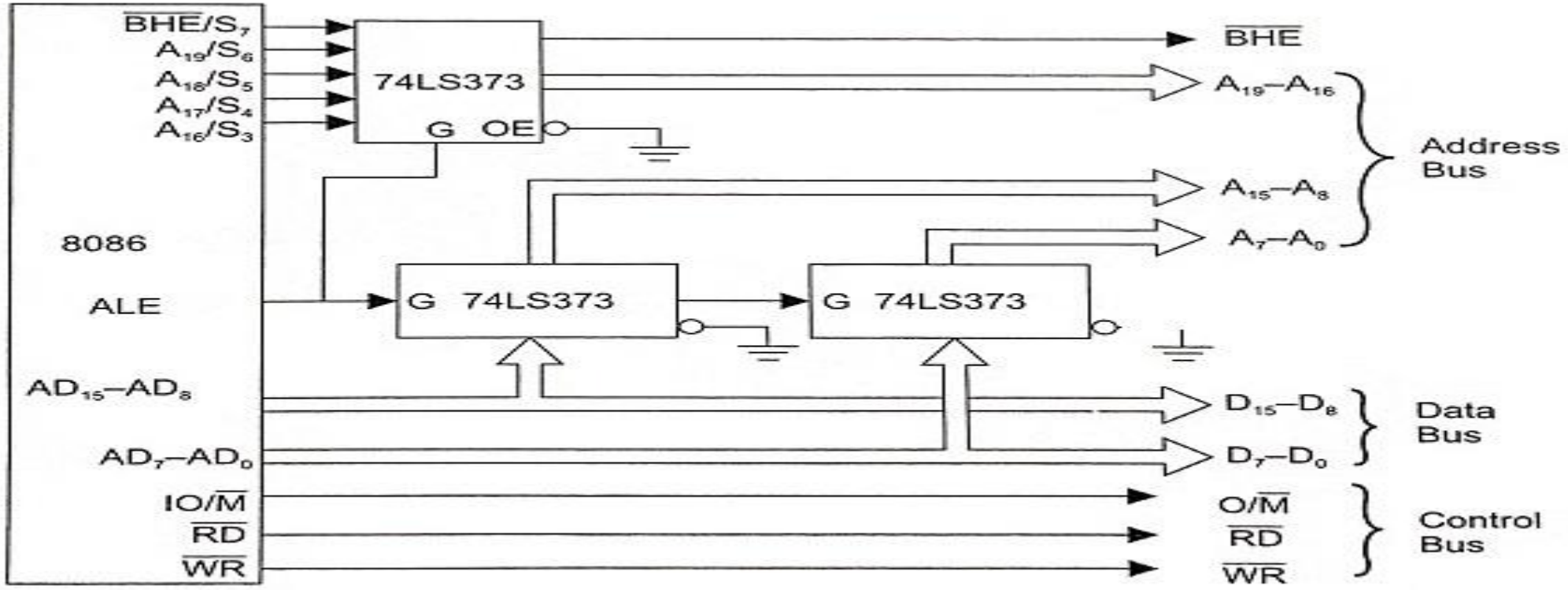


Fig. 5.28 *Demultiplexing of 20-bit address bus in 8086 processor*

control bus

- 3 pins for control bus
- Read RD
- Write W
- Input,output/memory IO/$\overline{M}$

0	1	0	
1	0	0	
0	1	1	
1	0	1	
0	0	X	never

Bus timing

- Multiplex： 8088 multiplex the lower 8 bits of the address bus and the data bus on AD0-AD7
- During the first clock cycle T1： AD carry the address
- T2,T3： are used for data transfer (read or write)
- T4： Data is sampled and read into the processor

- Ready→ connects to 8088 for synchronization

0(not ready(wait state)) , 1(ready for the data transfer)

- DEN→ Enable data and specify the direction flow of the data

For example: A(microprocessor) $\rightleftarrows$ B(memory)

- DT/ $\bar{R}$ → data transmit / receive

0(receive) , 1(transmit)

Sheet

- 1. State the main differences between the 8088 and 8086 pinouts. Are the two chips interchangeable?
- 2. ALE is an _____ (input, output) signal for the 8088.
- 3. What is the maximum number of bytes of memory addressable by the 8088, and why?
- 4. RESET is an _____ (input, output) signal for the 8088.
- 5. When the 8088 uses the pins for addresses, they are _____(input, output, both in and out), but when the

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- 1.8088 8bit data pins 8086 16bit data pins
- 2.Output
- 3.2^{20} = 1MB
- 4.Input
- 5.Output, both

Sheet

- 6. To use a math coprocessor with the 8088, one must connect the 8088 in _____ (maximum, minimum) mode.
- 7. True or false. An address must be latched from pins AD0–AD7 in the 8088.
- 8. Which of the following signals is provided by the 8088 CPU in minimum mode?

(a) INTR (b) ALE (c) WR (d) IO/M

- 9. What is the advantage of demultiplexing address/data in the 8088 CPU?
- 11. ALE is activated in which Tstate?
- 12. Why are 8086-based systems more expensive compared to 8088-based systems?
- 13. To use the 8088 with the 8087 math coprocessor, is the minimum/maximum pin connected to low or high?

Sheet

- 6.Maximum
- 7.true
- 8. a,b,c,d
- 9. Reduce the number of pins
- 11. T1
- 12. 8086 peripheral devices’ architecture must be completely changed (at that time) to meet with the 16-bit output data bus of the 8086
- 13.low

Sheet

Use the following for the next three problems.

MEMR* MEMW*

(a) 0 0

(b) 0 1

(c) 1 0

(d) 1 1

* Active low.

- 15. Which of the above control signals is activated during the memory read cycle?
- 16. Which one is activated during memory write?
- 17. Which of the above absolutely cannot happen at the same time?

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15. b

16. c

17. a